Arista Access Node HEB-ACC-51 Routing Protocol Flap Failure SOP

Node: HEB-ACC-51 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1472

Description

This Standard Operating Procedure is designed to address routing protocol flap failures on the Arista Access Node HEB-ACC-51 located at the Hebbal Edge site. The failure domain is characterized by a BGP route churn, OSPF adjacency loss, or ISIS LSP flooding, resulting in an ALARM_1472 warning. The underlying cause of this issue lies in the dynamic nature of routing protocols, which can sometimes lead to temporary instability and flap failures. The Arista network devices, specifically in the Access role, are configured to operate in a high-availability environment, with multiple links and routing protocols enabled to ensure seamless connectivity. However, when a failure occurs, the device's control plane may become overwhelmed, leading to packet loss, routing instability, and eventually, the ALARM_1472 warning. To mitigate this issue, it is essential to identify and resolve the root cause of the routing protocol flap failure, ensuring that the network infrastructure remains stable and available.

Pre-requisite Checks

- Pre-requisite: Verify that all redundant links are functioning correctly and that the Hebbal Edge site is accessible via a secondary path.

- Pre-requisite: Check the environmental conditions at the Hebbal Edge site, including temperature, humidity, and power supply, to ensure they are within the recommended operating parameters.

- Pre-requisite: Dispatch remote hands to the Hebbal Edge site to assist with troubleshooting and potential hardware swaps.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured routing policies or incorrect BGP neighbor settings.	High
OSPF adjacency loss due to network partitioning or incorrect OSPF configuration.	Medium
ISIS LSP flooding due to hardware or software issues with the Arista device.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, local router ID is 192.168.1.1, Status codes: s suppressed, d damped, h history, * valid, > best, i -internal, r RIB-failure, S Stale, Description: This command displays a summary of the BGP routing table, including the number of routes, peers, and any errors. It helps identify potential issues with BGP route chirns or misconfigured neighbor settings.
```

```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.2.2 1 Full/ - 00:00:32 192.168.2.2
Ethernet1, Description: This command displays a list of OSPF
neighbors, including their status, dead time, and interface
information. It helps identify potential issues with OSPF adjacency
loss or network partitioning.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID Seq Age LSP ID Seq Age 192.168.1.1 0x00000001 00:00:10 192.168.2.2
0x00000001 00:00:09, Description: This command displays the ISIS LSP
database, including LSP ID, sequence number, and age. It helps
identify potential issues with ISIS LSP flooding or hardware issues
with the Arista device.
```

Resolution Steps

Step 1: Step: Verify BGP routing policies and neighbor settings, Command: show bgp policy, Description: Verify that the BGP routing policies and neighbor settings are correctly configured and not causing the route churn.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf neighbor 192.168.2.2, Description: Reset the OSPF adjacency with the neighboring router at 192.168.2.2 to resolve the adjacency loss.

Step 3: Step: Clear ISIS LSP cache, Command: clear isis lsp-cache, Description: Clear the ISIS LSP cache to resolve the LSP flooding issue.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 1,
local router ID is 192.168.1.1, Status codes: s suppressed, d damped,
h history, * valid, > best, i -internal, r RIB-failure, S Stale,
Description: Verify that the BGP routing table is stable and there are
no errors.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.2.2 1 Full/ - 00:00:32 192.168.2.2
Ethernet1, Description: Verify that the OSPF adjacency is stable and
there are no errors.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID Seq Age LSP ID Seq Age 192.168.1.1 0x00000001 00:00:10 192.168.2.2
0x00000001 00:00:09, Description: Verify that the ISIS LSP database is
stable and there are no errors.
```

Roll-back Procedures

- Step: Revert BGP routing policies and neighbor settings, Command: copy run start, Description: Revert the BGP routing policies and neighbor settings to their previous configurations if the fix fails.

- Step: Reset OSPF adjacency, Command: clear ospf neighbor 192.168.2.2, Description: Reset the OSPF adjacency with the neighboring router at 192.168.2.2 to resolve the adjacency loss.

- Step: Clear ISIS LSP cache, Command: clear isis lsp-cache, Description: Clear the ISIS LSP cache to resolve the LSP flooding issue.